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Research Article

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Intrinsic Capacity Trajectories: Implications for Subsequent Falls and Hospitalizations Among Older Adults

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Abstract

Background: Intrinsic capacity (IC) is the composite of an individual's physical and mental capacities. However, the association between IC trajectories and falls and hospitalizations remains uncertain. This study aimed to determine the IC trajectories among older adults, investigating its association with subsequent risk of falls and hospitalizations.

Methods: This study enrolled 3 902 older adults aged ≥65 from the National Health and Aging Trends Study (Wave 2015–2019). A bifactor model was used for repeated measurements of the 5 IC domains to generate IC scores for 4 time points (Wave 2015–2018). IC trajectories were identified using group-based trajectory modeling, and modified Poisson regression was used to analyze the associations between IC trajectories and subsequent fall and hospitalization risk.

Results: The mean age of the participants was 76.70 years (standard deviation = 6.78), and the majority were female (57.3%). Three IC trajectories were identified, including persistently low (17.86%), persistently moderate (33.96%), and persistently high (48.18%). Compared with the persistently low class, the moderate and high classes have significantly lower fall and hospitalization risks. Multivariate-adjusted rate ratios fall occurrence were 0.87 (95% confidence interval [CI]: 0.78–0.98) and 0.74 (95% CI: 0.65–0.85), for multiple falls were 0.81 (95% CI: 0.68–0.96) and 0.52 (95% CI: 0.41–0.66), for hospitalization occurrence were 0.76 (95% CI: 0.66–0.87) and 0.48 (95% CI: 0.39–0.58), and for multiple hospitalizations were 0.65 (95% CI: 0.53–0.80) and 0.37 (95% CI: 0.28–0.48), respectively.

Conclusions: IC trajectories were associated with falls and hospitalizations. Strategies focusing on improving and maintaining IC at a higher level over time could help reduce the subsequent risk of falls and hospitalizations.

Keywords: Epidemiology, Healthcare, Injury prevention, Successful aging

Globally, the trend of population aging is rapidly accelerating, becoming a worldwide societal challenge. In response to this trend, the World Health Organization proposed the concept of "healthy aging" in 2015, emphasizing that maintaining the intrinsic capacity (IC) of older adults is crucial for achieving sustainable development in aging societies (1). IC, the composite of all the physical and mental capacities an individual can draw on, is considered a core element in maintaining the health and autonomy of older adults (2). It is typically assessed through a comprehensive scoring system based on 5 domains: cognitive capacity, locomotor capacity, sensory, vitality, and psychological capacity (3).

IC is a composite indicator, and studies have found that its component domains are associated with falls and hospitalizations. Although current multifactorial interventions have been implemented to address diverse risk factors, their effectiveness in reducing the risk of falls and hospitalizations among older adults has been limited (4). In such context, the "World guidelines for fall prevention and management for older adults" suggest that enhancing domains of IC, such as physical activity, sensory function, cognitive function, and nutritional status, can contribute to the prevention and management of falls and hospitalizations (4,5). Some research conducted in the community and nursing homes has suggested that IC can effectively predict adverse outcomes, for example, falls, disability, frailty, hospitalizations, and mortality (6–8). However, most of these studies on falls and hospitalizations have been cross-sectional, or the older population has been chiefly from hospitals or nursing homes with small and unrepresentative sample sizes (9,10). Additionally, current research mainly focused on the static aspects of IC, overlooking the fact that individuals may not experience the same rate and timing of decline (1)). In contrast, the IC of older adults tends to be a dynamic process (eg, remaining high or declining) over time (11). To our knowledge, only a few studies have examined IC trajectories and their impact on adverse outcomes, with no identified research specifically related to falls and hospitalizations (12,13). Falls and hospitalizations are 2 of the most common and essential acute adverse outcomes among older adults. Although falls can lead to injuries and even hospitalization, the prevention strategies for both may differ (14). Examining the independent effects of IC on both can provide a more comprehensive assessment of the combined risk of IC

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on the health of older adults and provide a scientific basis for developing preventive measures for different outcomes.

Using a cohort from the Wave 2015–2019 of the National Health and Aging Trends Study (NHATS), this study aims to (a) synthesize IC scores among older adults for each year from Waves 2015 to 2018; (b) identify the trajectory groups of IC over the previous 4-year span, and (c) to explore the associations between IC trajectories and subsequent 1-year risk of falls and hospitalizations.

Method

Study Population and Data Collection

The data for this study were obtained from the NHATS, a large-scale, nationally representative sample of Medicare beneficiaries aged 65 and older in the United States (including community, nursing home, and residential care facilities) (15). Sponsored by the National Institute on Aging (grant number NIA U01AG032947) and launched in 2011, NHATS conducts annual face-to-face interviews to gain in-depth insights into older adults' health, activities, and healthcare status. The Johns Hopkins University Institutional Review Board approved the NHATS protocol, and all participants provided written informed consent. This study used five waves of data (Wave 2015–2019) from the latest 2015 cohort, starting with 8 334 older adults participating in the survey. The inclusion criteria for our study cohort comprised (a) all participants aged 65 and above, (b) complete sex and age information, (c) participants in Wave 2015–2019 follow-ups, (d) complete information about IC in Wave 2015–2018 and complete information about falls and hospitalizations in Wave 2019. Ultimately, this study was narrowed down to 3 902 participants for analysis (Figure 1).

Intrinsic Capacity

Based on the WHO's conceptual framework for IC and current literature (3,16), we integrated the 5 domains of cognitive capacity, locomotor capacity, sensory, vitality, and psychological capacity into a composite IC score. IC was assessed on Wave 2015–2018, with a total of 4 waves (ie, 2015, 2016, 2017, and 2018), which were used to identify subsequent trajectory groups. The detailed description of each domain is shown in Supplementary Table 1.

Cognitive Capacity

Cognitive capacity was assessed through 3 tasks: memory (immediate and delayed 10-word recall), orientation (date, month, year, day of the week, naming President and Vice President), and executive function (clock drawing test) (17).

Locomotor Capacity

The Short Physical Performance Battery was used to assess locomotor capacity (18). It comprised 3 components: a hierarchical standing balance assessment, a 3-m gait speed evaluation, and 5 timed repetitive chair stand tests. Each part scores ranging from 0 to 4, with higher scores indicating better locomotor capacity.

Sensory

Hearing and vision impairments were measured using self-report (7). Hearing status was assessed by asking, "In the last month, have you used a hearing aid or other hearing device?" For vision, participants were also asked, "Do you wear glasses or contacts to help you see things?" Those who need hearing aids or deafness were determined to have poor hearing, and those who need glasses or blindness were determined to have poor vision.

Vitality

Vitality capacity was assessed by the grip strength and the peak air flow (7). Grip strength tests measured absolute grip strength (kg) using a digital, adjustable hand dynamometer (Jamar Plus Hydraulic Hand Dynamometer, USA). Each hand had 2 measurements, and the maximum grip strength value was selected for analysis. Peak airflow (L/min) was measured using a spirometer. Eligible participants were asked to take a deep breath and blow into the spirometer as hard as possible. They were then required to repeat the procedure to give two technically satisfactory blows. The highest technically satisfactory measure of peak airflow was used in the analysis (19).

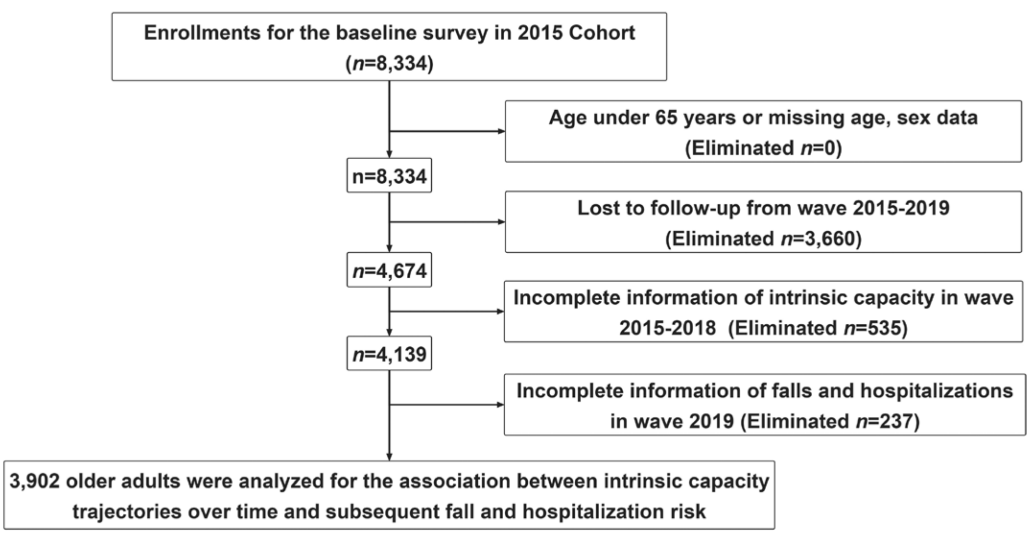

Figure 1. Flowchart of participant selection.

Psychological Capacity

Psychological capacity (depression and anxiety) was assessed using the Personal Health Questionnaire-4 (PHQ-4) scale (20). Each item corresponded with 4 categorized answers: not at all, several days, more than half the days, and nearly every day. The total scores for depression and anxiety were 0–6, respectively, with higher scores indicating lower psychological capacity.

Outcomes

Falling down in the NHATS questionnaire means any fall, slip, or trip in which you lose your balance and land on the floor or ground or at a lower level. The fall outcome variables were the fall occurrence and multiple falls, which were collected based on two questions in Wave 2019: "Since last year, have you fallen down?" and "Since last year, have you fallen down more than one time?" Fall occurrence indicated whether an individual experienced at least 1 fall (≥1), whereas multiple falls indicated whether an individual experienced more than one fall (≥2).

Hospitalization means a time when you stay at least 1 night in the hospital. The hospitalization outcome variables were hospitalization occurrence and multiple hospitalizations, including 2 questions in Wave 2019: "Have you had an overnight hospital stay since the time of the last interview in last year?" and "How many separate overnight hospital stays have you had since the time of the last interview in last year?" We defined hospitalization occurrence ≥2 as multiple hospitalizations.

Covariates

We included the following baseline variables as covariates: age, sex, marital status (married and not married), education (high school and below, beyond high school), income (poor, fair, good, very good), residence (metropolitan and non-metropolitan), self-reported health (good and poor), fall or hospitalization history, number of chronic diseases, and body mass index (BMI). The number of chronic diseases includes 10 chronic diseases such as heart disease, hypertension, arthritis, and osteoporosis. BMI was calculated as weight in kilograms divided by height squared in meters (kg/m^2).

Statistical Analysis

For characteristics description, mean ± standard deviation [*SD*] or the median with the interquartile range was used for continuous variables and numbers (percentage) for categorical variables. Overall differences between groups were tested using analysis of variance (ANOVA), Kruskal–Wallis test, or chi-square test.

The most recommended method, that is, the bifactor model, was used to compose the IC scores (7,21,22). To ensure the factor structure's applicability across all time points, a multigroup confirmatory factor analysis was conducted. The IC scores were standardized with a mean of 0 and an *SD* of 1. A positive IC score indicates above the mean, while a negative score indicates below it. Based on the established invariance, factor loadings and intercepts obtained from the baseline data (Wave 2015) were applied to generate IC scores for each subsequent time point. Subsequently, this study employed group-based trajectory modeling (GBTM) to identify 4-year (Wave 2015–2018) trajectories of IC among older adults. GBTM clusters individuals with similar trajectories into potential trajectory groups. Based on prior literature on IC trajectories among older adults (13), we hypothesized that there would be 2–6 distinct trajectories of IC. Then, we fitted and compared models with different numbers of groups and different trajectory shapes (linear, quadratic, and cubic). Determination of the optimal model relies on (a) minimizing Bayesian information criterion (BIC) and Akaike's information criterion (AIC) values; (b) ensuring each individual's average posterior probability of belonging to a specific trajectory group exceeds 70%; (c) maintaining a membership proportion of 5% or more for each trajectory group; (d) ensuring the model exhibits sufficient clinical relevance and interpretability (23). Ultimately, a 3-class model was determined to be the optimal choice. Subsequently, the ordinal logistic regression model was used to analyze the determinants of IC trajectory groups and modified Poisson regression models without and with adjustment for covariates to analyze the association between IC trajectories and four adverse outcomes: fall occurrence, multiple falls, hospitalization occurrence, and multiple hospitalizations (24). Models were adjusted for age, gender, marital status, education, income, residence, self-reported health, fall or hospitalization history, number of chronic diseases, and BMI. The adjusted odds ratio (OR), rate ratio (RR), and 95% confidence intervals (CIs) were reported. Subsequently, 2 sensitivity analyses were performed. First, we use another simple and commonly used method for generating IC scores, that is, the *Z*-score method (25). A composite IC *Z*-score was constructed as the sum of the individual *Z*-scores of cognitive capacity, locomotor capacity, sensory, vitality, and psychological capacity domains divided by 5 to create the final IC composite *Z*-scores from Waves 2015 to 2018, respectively. The cognitive function *Z*-score was obtained by averaging the *Z*-score for memory, orientation, and executive function. Similarly, locomotor capacity was obtained by averaging the *Z*-scores for balance, gait speed, and chair stand. A sensory was derived from the average of hearing *Z*-scores and vision *Z*-scores, while a vitality *Z*-score was computed based on the average of grip strength *Z*-scores and peak air flow *Z*-scores. Notably, the positive sensory and psychological domain *Z*-scores were converted to a negative value to create an index reflecting that the higher scores indicate better IC. Second, given that IC decreases with age, we also used the GBTM to identify distinct IC trajectories with age and analyzed the association between the IC trajectories with age and adverse outcomes.

The bifactor model (lavaan package), descriptive analysis, ordinal logistic regression (MASS package), and modified Poisson regression with robust (sandwich package) were fitted using R 4.3.0. The GBTM was conducted using the "Traj" program in Stata version 16.0 (StataCorp LLC, College Station, TX). A 2-sided *p*-value <.05 was considered statistically significant.

Results

A total of 3 902 older adults aged ≥65 participated in all 5 waves of the study (Table 1). Among them, the mean age was 76.70 years (*SD* = 6.78), and the majority were female (57.3%). A bifactor model was applied to generate IC scores from Wave 2015 to 2018, showing a good fit to the data with the root-mean-square error of approximation of 0.033 (90% CI: 0.029–0.037), the comparative fit index of 0.980, the Tucker–Lewis index of 0.970, and the standardized

Table 1. Baseline Characteristics of Participants by Intrinsic Capacity (IC) Trajectories

Variables	Total (n = 3 902)	Trajectory Class			p-Value
		Class 1 (n = 697)	Class 2 (n = 1 325)	Class 3 (n = 1 880)	
Sociodemographic factors					
Age (years)	76.70 (6.78)	80.88 (7.28)	78.12 (6.63)	74.14 (5.51)	<.001
Gender					<.001
Male	1 665 (42.7%)	156 (22.4%)	469 (35.4%)	1 040 (55.3%)	
Female	2 237 (57.3%)	541 (77.6%)	856 (64.6%)	840 (44.7%)	
Marital status					<.001
Married	1 830 (46.9%)	165 (23.7%)	512 (38.6%)	1 153 (61.3%)	
Not married	2 072 (53.1%)	532 (76.3%)	813 (61.4%)	727 (38.7%)	
Education					<.001
High school and below	1 688 (43.3%)	467 (67.0%)	681 (51.4%)	540 (28.7%)	
Beyond high school	2 214 (56.7%)	230 (33.0%)	644 (48.6%)	1 340 (71.3%)	
Income					<.001
Poor	1 014 (26.0%)	386 (55.4%)	441 (33.3%)	187 (9.9%)	
Fair	937 (24.0%)	172 (24.7%)	397 (30.0%)	368 (19.6%)	
Good	979 (25.1%)	93 (13.3%)	292 (22.0%)	594 (31.6%)	
Very good	972 (24.9%)	46 (6.6%)	195 (14.7%)	731 (38.9%)	
Residence					.410
Metropolitan	3 157 (80.9%)	563 (80.8%)	1 087 (82.0%)	1 507 (80.2%)	
Non-metropolitan	745 (19.1%)	134 (19.2%)	238 (18.0%)	373 (19.8%)	
Health conditions					
Self-reported health					<.001
Good	3 169 (81.2%)	382 (54.8%)	1 034 (78.0%)	1 753 (93.2%)	
Poor	733 (18.8%)	315 (45.2%)	291 (22.0%)	127 (6.8%)	
Fall history					<.001
Yes	1 241 (31.8%)	311 (44.6%)	439 (33.1%)	491 (26.1%)	
No	2 661 (68.2%)	386 (55.4%)	886 (66.9%)	1 389 (73.9%)	
Hospitalization history					<.001
Yes	674 (17.3%)	193 (27.7%)	254 (19.2%)	227 (12.1%)	
No	3 228 (82.7%)	504 (72.3%)	1 071 (80.8%)	1 653 (87.9%)	
Number of chronic diseases	2.0 (1.0, 3.0)	3.0 (2.0, 4.0)	2.0 (1.0, 3.0)	2.0 (1.0, 2.0)	<.001
BMI (kg/m^2)	27.95 (5.61)	28.43 (6.69)	28.08 (5.98)	27.68 (4.83)	.006

Note: BMI = body mass index.

root-mean-square residual (SRMR) of 0.024 (Supplementary Figure 1). The results of longitudinal measurement invariance indicated that the more restrictive model demonstrated measurement invariance across the time points (Supplementary Table 2). Three distinct IC trajectories over time were identified based on 4-wave IC scores (Supplementary Tables 3 and 4). Specifically, the first class (Class 1) was identified as "persistently low" (n = 697, 17.86%), that is, low baseline IC with a significantly declining trajectory. The second class (Class 2) was identified as "persistently moderate" (n = 1 ,325, 33.96%), that is, moderate baseline IC with a slightly declining trajectory. The third class (Class 3) was identified as "persistently high" (n = 1 880, 48.18%), that is, high baseline IC with a slightly declining trajectory (Figure 2). In an ordinal logistic regression analysis, beyond high school, fair income or good income or very good income, non-metropolitan residence, non-fall history, and non-hospitalization history were associated with a high level of IC trajectory class (Class 3), whereas age, female, poor health, number of chronic diseases, and BMI were associated with a lower level of IC trajectory class (Class 1; Supplementary Table 5).

Table 2 shows the associations between IC trajectories over time and subsequent risk of falls and hospitalizations. Compared with Class 1, Class 2 and Class 3 have a significantly lower risk of fall occurrence, multiple falls, hospitalization occurrence, and multiple hospitalizations (all p-value <.001). Among these, the multivariate-adjusted RRs for fall occurrence were 0.87 (95% CI: 0.78–0.98) and 0.74 (95% CI: 0.65–0.85), for multiple falls were 0.81 (95% CI: 0.68–0.96) and 0.52 (95% CI: 0.41–0.66), and for hospitalization occurrence were 0.76 (95% CI: 0.66–0.87) and 0.48 (95% CI: 0.39–0.58), and the adjusted RRs for multiple hospitalizations were 0.65 (95% CI: 0.53–0.80) and 0.37 (95% CI: 0.28–0.48), respectively.

Sensitivity Analysis

Three distinct IC trajectories with age were identified (Supplementary Tables 6 and 7), and all 3 trajectories have shown a declining trend with age (Supplementary Figure 2). Similarly,

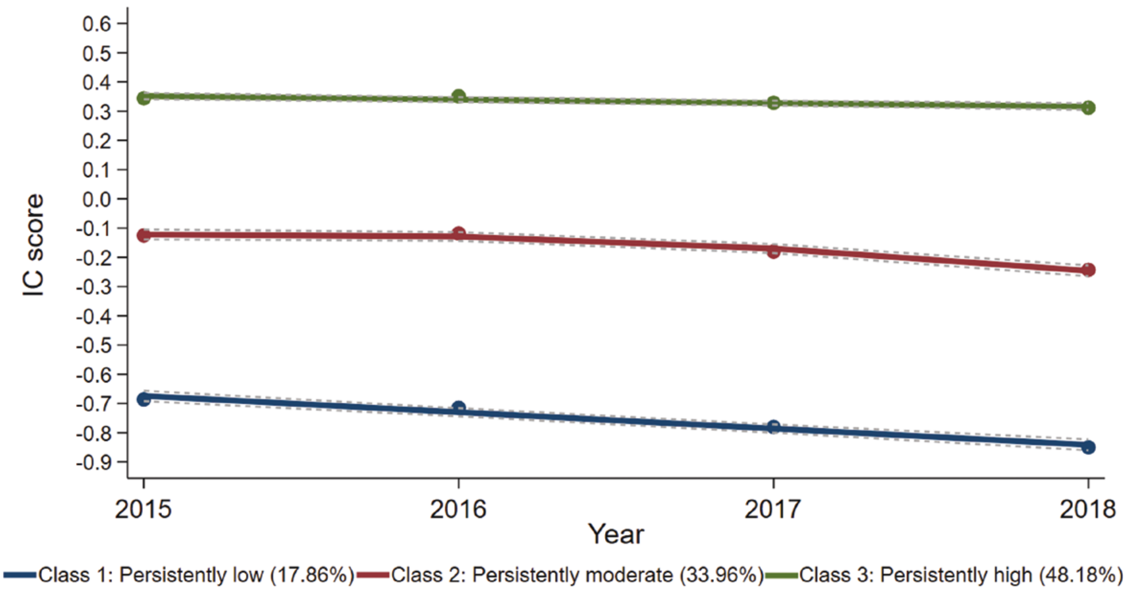

Figure 2. Intrinsic capacity (IC) trajectories over time among older adults. The solid line represents the estimated mean, and the dashed line represents the 95% confidence interval of the mean.

Table 2. Associations Between Intrinsic Capacity (IC) Trajectories Over Time With Subsequent Falls and Hospitalizations (*n* = 3 902)

IC trajectories	Model 1* RR (95% CI)				Model 2† Adj. RR (95% CI)			
	Fall occurrence	Multiple falls	Hospitalization occurrence	Multiple hospitalizations	Fall occurrence	Multiple falls	Hospitalization occurrence	Multiple hospitalizations
Class 1 (reference)	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00
Class 2	**0.81 (0.73, 0.91)**	**0.71 (0.59, 0.84)**	**0.65 (0.57, 0.74)**	**0.56 (0.47, 0.68)**	**0.87 (0.78, 0.98)**	**0.81 (0.68, 0.96)**	**0.76 (0.66, 0.87)**	**0.65 (0.53, 0.80)**
Class 3	**0.63 (0.56, 0.71)**	**0.37 (0.31, 0.46)**	**0.35 (0.30, 0.40)**	**0.28 (0.23, 0.34)**	**0.74 (0.65, 0.85)**	**0.52 (0.41, 0.66)**	**0.48 (0.39, 0.58)**	**0.37 (0.28, 0.48)**

Notes: Adj. RR = adjusted rate ratio; BMI = body mass index; CI = confidence interval. The values in bold indicate statistically significant results ($p < .05$).
*Model 1 estimated unadjusted rate ratio from modified Poisson regression models.
†Model 2 was adjusted for age, gender, marital status, education, income, residence, self-reported health, fall or hospitalization history, number of chronic diseases, and BMI.

3 distinct IC trajectories over time (*Z*-score) were identified (Supplementary Tables 8 and 9 and Supplementary Figure 3). Additionally, we also found similar results for IC trajectories with age or over time (*Z*-score) associated with the risk of fall occurrence, multiple falls, hospitalization occurrence, and multiple hospitalizations (Supplementary Tables 10 and 11).

Discussion

Based on data from Waves 2015 to 2018 (4 waves) in NHATS, IC scores were generated from the 5 domains using a bifactor model, which yielded a good fit to the data and supported the construction of IC as an integral latent variable. Three IC trajectories were identified through the 4 waves of IC scores generated by the bifactor model using the GBTM, that is, persistently low (17.86%), persistently moderate (33.96%), and persistently high (48.18%), with persistently low scores showing a more pronounced decline trend than the other 2 classes. Compared with the persistently low class, the moderate and high classes have significantly lower risks of fall occurrence, multiple falls, hospitalization occurrence, and multiple hospitalizations.

Given the current lack of clear diagnostic criteria for IC impairment, monitoring IC in older adults enables the capture of long-term trends while considering population heterogeneity. This approach also contributes to a deeper understanding of the potential mechanisms underlying adverse outcomes, facilitating precise interventions (16). Our results found 3 classes of trajectories, which is consistent with other previous studies. For example, Yu et al. found 3 similar classes of trajectories in a 4-year IC analysis of 1 371 community-dwelling older adults aged 60 years and older (13), and another IC trajectory analysis from 2011 to 2015 found that 3 893 older adults aged 60 years and older remained on a 3-category similar trajectory (26). Interestingly, the final IC trajectory results were close despite differences in the populations, follow-up durations, and IC synthesis methods used in different studies. Salinas-Rodríguez et al. analyzed trajectories of 9 years of IC in 2 735 adults aged 50 years and older. They found 3 trajectories (ie, steeply declining, moderately declining, and slightly increasing) (27). Possible reasons for the more pronounced trajectories were using a 0–100 point transformed method and a more extended follow-up period (9 years).

We found that trajectories of IC in older adults were associated with the risk of subsequent fall occurrence and multiple falls. Currently, only limited studies have examined the association between IC decline and falls among older adults. However, the IC was captured at a single point in time. For example, Shen et al.'s study of 703 hospitalized patients found that declined IC composite scores were associated with increased risks of falls (OR = 0.64) (28). Charles et al.'s study of 604 nursing home residents found that the risk of falling decreased when there was a 1-unit increase in balance performance (hazard ratio [HR] = 0.87) and nutrition score (HR = 0.96). However, no association was found between IC and repeated falls (6). Another study of 756 community-dwelling older adults found that only visual impairment in IC predicted recurrent falls (10). A possible reason for low IC leading to fall occurrence or multiple falls is that all domains of IC are common risk factors for falls, especially the locomotor and psychological domains (29,30).

In terms of hospitalization occurrence and multiple hospitalizations, current studies are less reported. Decreased IC in older adults may be associated with deteriorating health conditions, for example, chronic diseases (31), dysfunction (21), or acute medical conditions (32), which may require hospitalization. Ramírez-Vélez et al. found that low IC was associated with increased cardiovascular disease (CVD) incidence and its mortality in middle-aged and older adults (33). Another study found that a 1-point lower IC value was associated with a 7% increase in the risk of activities of daily living disability, a 6% increase in the risk of a nursing home stay, and a 5% increase in mortality (34). On the other hand, decreased IC may be associated with balance and coordination problems, increasing the risk of falls and fractures in older adults, which may result in the need for hospitalization (35).

The strength of this study is that, to the authors' knowledge, it is the first study that dynamically examined IC trajectories over time and prospectively assessed their association with subsequent risk of falls and hospitalizations. However, several limitations also required attention. First, the follow-up duration was relatively short (4 years), which led to the inability to observe long-term dynamic changes in IC. Second, some domains of IC (eg, sensory) were self-reported, making measurement errors inevitable. Third, due to data availability, the vitality domain was evaluated using grip strength and peak airflow, but these are also commonly used and recognized indicators in many studies.

There are advantages and disadvantages to using the bifactor model and directly averaging the Z-scores of the 5 domains to generate IC scores. The bifactor model reveals the contribution of the domains to IC through detailed structural analyses (especially when measurements of each domain are not uniform), but this method is characterized by complexity and high data quality requirements (36). In contrast, the direct average z-score approach is easy to operate and suitable for rapid analysis and interpretation of results, but ignores the relationships and differences in contributions among domains (37). Future studies could compare the predictive performance of different IC synthesis methods for adverse outcomes to find the most appropriate strategy.

In conclusion, this study suggests that IC trajectories among older adults were associated with falls and hospitalizations. This implies that monitoring IC among older adults indeed has its clinical utility. Strategies focusing on improving and maintaining IC at a higher level over time could help reduce the risk of subsequent falls and hospitalizations.

Supplementary Material

Supplementary data are available at *The Journals of Gerontology, Series A: Biological Sciences and Medical Sciences* online.

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Conflict of Interest

None.

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